Notation and Basic Concepts

Bras and Kets

A (pure) quantum state is denoted by a ket $|\psi\rangle \in \mathcal{H}$. Its dual vector is the bra $\langle\psi| \in \mathcal{H}^*$, and the inner product, written as $\langle\phi|\psi\rangle$, is a complex number. Kets can be thought of as column vectors, while bras are row vectors (the Hermitian conjugate of the ket). The operation that takes a ket to its corresponding bra is the Hermitian conjugate and it is an *antilinear* map:

$$\langle a\psi| = a^*\langle\psi| = |a\psi\rangle^\dagger = (a|\psi\rangle)^\dagger, \tag{1}$$

By antilinearity, we mean that complex scalars are conjugated when pulled out of the bra. Bras can also be thought of as linear maps from kets to complex numbers:

$$\langle\phi| : \mathcal{H} \to \mathbb{C}, \qquad |\psi\rangle \mapsto \langle\phi|\psi\rangle. \tag{2}$$

As a *linear* map, bras act linearly on kets:

$$\langle\phi|(a|\psi_1\rangle + b|\psi_2\rangle) = a\langle\phi|\psi_1\rangle + b\langle\phi|\psi_2\rangle, \tag{3}$$

However, bras are *antilinear* in their own argument:

$$\langle a\phi_1 + b\phi_2|\psi\rangle = a^*\langle\phi_1|\psi\rangle + b^*\langle\phi_2|\psi\rangle. \tag{4}$$

A state is normalized if $\langle\psi|\psi\rangle = 1$. Two states $|\phi\rangle$ and $|\psi\rangle$ are orthogonal if $\langle\phi|\psi\rangle = 0$. Finally, the outer product $|\phi\rangle\langle\psi|$ is an operator that maps kets to kets:

$$|\chi\rangle \mapsto |\phi\rangle\langle\psi|\chi\rangle. \tag{5}$$

This sums up basically everything you need to know about bras and kets.

Bases

A complete basis $\{|i\rangle\}$ of the Hilbert space $\mathcal{H}$ satisfies the completeness relation

$$\sum_i |i\rangle\langle i| = \mathbb{1}, \tag{6}$$

where $\mathbb{1}$ is the identity operator on $\mathcal{H}$. If the basis is orthonormal, then

$$\langle i|j\rangle = \delta_{ij}. \tag{7}$$

For a continuous basis $\{|\alpha\rangle\}$, the dimension of the Hilbert space is infinite and the completeness relation becomes

$$\int d\alpha\, |\alpha\rangle\langle\alpha| = \mathbb{1}, \tag{8}$$

with orthonormality

$$\langle\alpha|\alpha'\rangle = \delta(\alpha - \alpha'). \tag{9}$$

In particular, the position basis $\{|\vec{x}\rangle\}$ satisfies

$$\int d^3x\, |\vec{x}\rangle\langle\vec{x}| = \mathbb{1}, \qquad \langle\vec{x}|\vec{x}'\rangle = \delta^{(3)}(\vec{x} - \vec{x}'). \tag{10}$$

Time in nonrelativistic quantum mechanics is treated as a parameter and not a part of our basis labels, operators, or observables.

Operators and Expectation Values

Physical quantities (observables) are represented by linear operators on $\mathcal{H}$, such as

$$\hat{O}, \qquad \hat{x}, \qquad \hat{p}. \tag{11}$$

Given a basis, these operators can have a matrix representation. For instance, if $\{|i\rangle\}$ is an orthonormal basis, then the matrix elements of $\hat{O}$ are

$$O_{ij} = \langle i|\hat{O}|j\rangle. \tag{12}$$

The action of $\hat{O}$ on a state $|\psi\rangle$ can be written by two uses of the completeness relation in Eq. (6) as

$$\hat{O}|\psi\rangle = \sum_{i,j} |i\rangle\, \langle i|\, \hat{O}\, |j\rangle\, \langle j|\psi\rangle = \sum_{i,j} |i\rangle O_{ij} \langle j|\psi\rangle. \tag{13}$$

Given a normalized state $|\psi\rangle$, the expectation value of an operator $\hat{O}$ is

$$\langle O\rangle_\psi \equiv \langle\psi|\hat{O}|\psi\rangle. \tag{14}$$

In the position basis, the probability density to find the particle near $\vec{x}$ at time t is

$$\rho(\vec{x},t) = |\psi(\vec{x},t)|^2, \qquad \int d^3x\, \rho(\vec{x},t) = 1. \tag{15}$$

Hilbert Space

The Hilbert space $\mathcal{H}$ can be finite or infinite-dimensional in nonrelativistic quantum mechanics. The need for infinite-dimensional ones can be linked to the canonical commutation relation:

$$[\hat{x},\hat{p}] = i\hbar\, \mathbb{1}. \tag{16}$$

If $\mathcal{H}$ were finite-dimensional, the trace would be well-defined and we would have, for any operators A, B,

$$\mathrm{Tr}([A,B]) = \mathrm{Tr}(AB) - \mathrm{Tr}(BA) = 0, \tag{17}$$

where we used the property of the trace that $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$. Note that the trace operation on the basis $\{|i\rangle\}$ is defined as

$$\mathrm{Tr}(A) = \sum_i \langle i|A|i\rangle. \tag{18}$$

But taking the trace of the canonical commutator would give

$$\mathrm{Tr}([\hat{x},\hat{p}]) = \mathrm{Tr}(i\hbar\,\mathbb{1}) = i\hbar\,\mathrm{Tr}(\mathbb{1}) \neq 0, \tag{19}$$

which is a contradiction in finite dimensions. A similar argument applies to the energy operator. The precise mathematical statement is given by the Stone-von Neumann theorem, sometimes referred to as Pauli's theorem.

State Decomposition and Wavefunctions

If $\{|i\rangle\}$ is a discrete orthonormal basis, any state can be expanded as

$$|\psi\rangle = \sum_i c_i\, |i\rangle, \qquad c_i = \langle i|\psi\rangle. \tag{20}$$

The corresponding bra is

$$\langle\psi| = \sum_i c_i^*\, \langle i|, \tag{21}$$

and normalization reads

$$\langle\psi|\psi\rangle = \sum_i |c_i|^2 = 1. \tag{22}$$

In a continuous basis, the "components" of a state become functions. For example, in one spatial dimension,

$$|\psi\rangle = \int dx\, \psi(x)\, |x\rangle, \qquad \psi(x) = \langle x|\psi\rangle. \tag{23}$$

Thus the wavefunction is simply a *coefficient* of the ket $|x\rangle$ in the expansion of $|\psi\rangle$. Coefficients in the expansion of our state vector are precisely what we need to compute expectation values and probabilities, and therefore, observables.

Using completeness, any state can be written as a superposition of position eigenstates:

$$|\psi\rangle = \int d^3x\, |\vec{x}\rangle\langle\vec{x}|\psi\rangle = \int d^3x\, |\vec{x}\rangle\, \psi(\vec{x}). \tag{24}$$

In the momentum basis, the wavefunction is usually represented by

$$\phi(\vec{p}) \equiv \langle\vec{p}|\psi\rangle, \qquad |\psi\rangle = \int d^3p\, |\vec{p}\rangle\, \phi(\vec{p}). \tag{25}$$

In Sakurai and Napolitano, the notation $|\alpha\rangle$ for some state defined by the label α has a position-space wavefunction

$$\psi_\alpha(\vec{x}) \equiv \langle\vec{x}|\alpha\rangle, \qquad |\alpha\rangle = \int d^3x\, |\vec{x}\rangle\, \psi_\alpha(\vec{x}). \tag{26}$$

We will make use of this notation when necessary. What we will not write is $|\psi(x)\rangle$ since this is extremely ambiguous: is x a label or a variable?

Observables and Operators

In the standard postulates, observables correspond to Hermitian (self-adjoint) operators. The key point is that Hermitian operators have real eigenvalues and a complete set of orthogonal eigenkets (under appropriate conditions).

> The proof is simple: let $\hat{O}$ be Hermitian and let $|\alpha\rangle$ be an eigenket, $\hat{O}|\alpha\rangle = \alpha|\alpha\rangle$. Taking the Hermitian conjugate gives $\langle\alpha|\hat{O}^\dagger = \langle\alpha|\hat{O} = \alpha^*\langle\alpha|$. Now, consider two generic eigenkets $|\alpha\rangle$ and $|\alpha'\rangle$. Dotting these onto the equations above, we get
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> $$\langle\alpha'|\left(\hat{O}\,|\alpha\rangle\right) = \alpha\,\langle\alpha'|\alpha\rangle \qquad \text{and} \qquad \left(\langle\alpha'|\,\hat{O}\right)|\alpha\rangle = \alpha'^*\,\langle\alpha'|\alpha\rangle. \tag{27}$$
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> Equating these two expressions, we get $(\alpha - \alpha'^*)\,\langle\alpha'|\alpha\rangle = 0$. Now, note that if we set $\alpha' = \alpha$, then $\alpha = \alpha^*$ (real eigenvalues), however, if $\alpha \neq \alpha'$ then $\langle\alpha'|\alpha\rangle = 0$ (orthogonality of distinct eigenkets).

For a discrete spectrum $\hat{O}|i\rangle = O_i|i\rangle$ and $|\psi\rangle = \sum_i c_i|i\rangle$,

$$\langle O\rangle = \langle\psi|\hat{O}|\psi\rangle = \sum_i |c_i|^2\, O_i. \tag{28}$$

For a continuous spectrum labeled by α with wavefunction $\psi(\alpha) = \langle\alpha|\psi\rangle$,

$$\langle O\rangle = \int d\alpha\, |\psi(\alpha)|^2\, \alpha. \tag{29}$$

Mathematical Identities

Here are some useful mathematical identities that you may need when solving problems.

Dirac delta function: The Delta function can be defined through its most important property:

$$\int_{-\infty}^{\infty} dx\, f(x)\, \delta(x-x') = f(x'). \tag{30}$$

Another useful representation is through the Fourier transform of the constant (unity) function:

$$\delta(x-x') = \frac{1}{2\pi}\int_{-\infty}^{\infty} dk\, e^{ik(x-x')}. \tag{31}$$

Other definitions include the limit of an infinitely narrow Gaussian function:

$$\delta(x-x') = \lim_{\sigma\to 0} \frac{1}{\sigma\sqrt{\pi}} e^{-(x-x')^2/\sigma^2}, \tag{32}$$

or the limit of a *sinc* function:

$$\delta(x-x') = \lim_{N\to\infty} \frac{\sin(N(x-x'))}{\pi(x-x')}. \tag{33}$$

Some useful properties include:

$$\delta(ax) = \frac{1}{|a|}\delta(x), \qquad a>0 \text{ and } a\in\Re, \tag{34}$$

$$\delta(f(x)) = \sum_i \frac{\delta(x-x_i)}{|f'(x_i)|}, \qquad f(x_i)=0, \tag{35}$$

$$\delta(x-x') = \delta(x'-x), \tag{36}$$

$$\frac{d}{dx}\theta(x-x') = \delta(x-x'), \tag{37}$$

where $\theta(x-x')$ is the Heaviside step function.

Gaussian integrals: The basic Gaussian integral is

$$\int_{-\infty}^{\infty} dx\, e^{-ax^2} = \sqrt{\frac{\pi}{a}}, \qquad a>0. \tag{38}$$

More generally,

$$\int_{-\infty}^{\infty} dx\, x^n e^{-ax^2} = \begin{cases} 0, & n \text{ odd}, \\ \frac{(n-1)!!}{2^{n/2}}\sqrt{\frac{\pi}{a^{n+1}}}, & n \text{ even}. \end{cases} \tag{39}$$

For Gaussian with linear and constant terms,

$$\int_{-\infty}^{\infty} dx\, e^{-(ax^2+bx+c)} = \sqrt{\frac{\pi}{a}} e^{\frac{b^2-4ac}{4a}}, \qquad a>0. \tag{40}$$

In multiple dimensions,

$$\int d^n x\, e^{-\mathbf{x}^T A \mathbf{x}} = \frac{\pi^{n/2}}{\sqrt{\det A}}, \qquad A = A^T > 0. \tag{41}$$